



...Use The Windows Recovery Console

A broken Windows installation doesn't necessarily mean a reinstall is required—try using the commands in the Windows Recovery Console first

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No operating system is immune to damage, and Microsoft OSes are certainly no exception—viruses and spyware don't help matters at all! Thankfully, Windows 2000 and XP make recovery possible in a few easy steps. If XP refuses to start up or run properly, or if the OS choice does not appear, for example, you can fix it in a matter of minutes. All you need is your bootable Windows 2000 / XP CD.

We suggest you do a dry run of the Windows Recovery Console and familiarise yourself with it after reading this, so you don't feel lost when you do try a recovery. The process is easy if you have some knowledge of DOS.

Bringing Up The Console

Set your first boot device to CD-ROM and boot using your Windows boot CD. When you see "Press any key to boot from CD", do so. Next, choose the repair or recovery option by pressing [R]. If you have more than one installation of Windows, they will appear in the choices menu, so be aware of where the copy of Windows you wish to repair is installed, so you can choose it correctly. You need to enter the Administrator password in order to "log on" to the Windows installation in question. (Bear in mind that if you type a wrong password thrice in succession, the computer restarts.)

You will see a command prompt that says "X:\WINDOWS", where X is the partition where you installed the operating system ("C" in most cases). You can now access the troubleshoot / repair commands. Type "help"



Driver Conflicts

If your Windows is not starting up due to a driver conflict and the programs are all fine, you can first replace just the "system" file and check if Windows starts up. If this does not work, then replace the "software" file as well.

and press [Enter] to see a list of available commands. There are several powerful commands available, and we will explain the more important of these.

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Microsoft Windows XP [i386] Recovery Console.
The Recovery Console provides system repair and recovery
functionality.
Type HELP to quit the Recovery Console and restart the computer.
C: >
C: >
Which Windows installation would you like to log onto
(the number, press ENTER)?
```

Following this is a plain DOS prompt

fixboot And fixmbr

When you are installing a new OS, or have accidentally deleted system files from your boot drive, these two commands come to your rescue. The "fixboot" command rewrites the boot sector code onto the chosen partition. The syntax for this command is "fixboot <drive letter>". The "fixmbr" command repairs the boot record of the boot partition. The master boot record can be damaged by a virus or a faulty installation of a second OS. Just use the "fixmbr" command to repair your boot record and have Windows working again!

If neither of the above commands can get your Windows installation started, the next step will be to try the "bootcfg" command. Type "bootcnf /rebuild" for the recovery console to display all available Windows partitions. Choose the one you want to add to the boot menu.

chkdsk

The Check Disk (chkdsk) command scans a partition for errors and tries a data recovery if possible. If a bad sector is encountered, the Check Disk tool marks them as bad such that data will not be written to it. If Windows refuses to start and you suspect a failing hard disk (screeching noises, sudden hangs, etc.), use this command to check the status of your hard disk.

The syntax is "chkdsk X", where X is the drive letter assigned to the partition in question.

Restoring The Registry

Windows XP stores its Registry information in two files named "software" and "system" in the Windows\system32\config folder. Soon after installation, a copy of these files is stored in the Windows\repair folder. In the event of Windows not starting up at all, a sure-shot way of getting it up and running is to replace the existing "software" and "system" files in the Windows\system32\config with the files from the

Windows\repair folder. Note that none of your programs or drivers will work—you need to reinstall each of them. While it may seem like a tedious job, it sure is quicker and easier than a full-fledged reinstall and data backup.

To do this, at the "C:\WINDOWS" prompt, type "attrib -h -s \system32\config\system" and press [Enter]; next, type "attrib -s -h \system32\config\software" and press [Enter]. This is to remove the "system" and "hidden" attributes of the files. You can now copy the files from the repair folder by typing "copy \repair\system \system32\config", pressing [Enter], then typing "copy \repair\software \system32\config" and pressing [Enter]. When you're prompted to proceed with an overwrite, press [Y].

Run It From The Hard Disk

You can install the recovery console to your hard disk so that you can access it without the need for the Windows install CD. This speeds up performance because reading from hard disk is way faster than from CD. This is also useful if both your Windows installation and your CD-ROM drive die at the same time.

To do this, insert your XP CD while Windows is running, go to **Start > Run**, and type "X:\i386\winnt32.exe /cmdcons" where X is the letter of your CD-ROM drive. Bear in mind that if the boot record or boot files on your hard disk are corrupted, you will still need the CD to access the recovery console. ■

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